

The AI Divide in the Global South: A Case Study of Rwanda

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Introduction

While the debate about artificial intelligence in the Global North turns on questions of overuse and risk, the more urgent question in the Global South is whether AI is reaching its populations at all. This paper examines the AI divide using Rwanda as a case study and the country's education system as its narrowed sector of focus. Drawing on literature and policy, the paper engages three questions: how AI access shapes educational outcomes in Rwanda compared to peer countries in the Global North; which policy instruments determine who gets access to frontier AI; and whether a Global-South-led strategy can realistically close the gap. All taken into account, evidence suggests the **gap** is real and layered, that it is governed by policy instruments at four levels of authority, and that meaningful closure depends not on full sovereignty or passive cooperation but on a hybrid model in which the Global-South party sets the terms.

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Oxford's 2024 Word of the Year was "brain rot," a shorthand in the Global North for the intellectual decay attributed to large language models (LLMs) and chronic use of social media platforms. On its face value, one might successfully claim that it is the net impact of abundance: "he whose belly is bloated suffers as much as he who is starving" African Proverb. However, the contradiction is only apparent. The brain rot of the North is a late-stage symptom of saturation, while saturation is itself a privilege. The artificial intelligence (AI) that rots minds in the North is not the same AI that could transform a Rwandan classroom. One is consumption; the other is leverage.

By a parallel contrast, the roughly 2.6 billion people still offline in the Global South¹ are not simply deprived of digital tools; they are increasingly deprived of what is rapidly becoming a basic need. For economies already on the red side of the poverty line and the digital divide, the cost of being left behind a second time is too expensive to afford.

AI Access and Educational Outcomes in Rwanda

Before any discussion of AI in Rwandan classrooms, it's only fair to floor down some critical facts of the case: The ITU (2024) reports that approximately 2.6 billion people, roughly 32 percent of the world's population, remained offline in 2024, with internet penetration at 93 percent in high-income countries and only 27 percent in low-income ones. Rwanda sits within this lower tier: 38 percent of the population was online by mid-2025, with a stark internal split between urban areas at 57 percent and rural areas at 19 percent. The topographic reach of this disparity is shown in Figure 1.

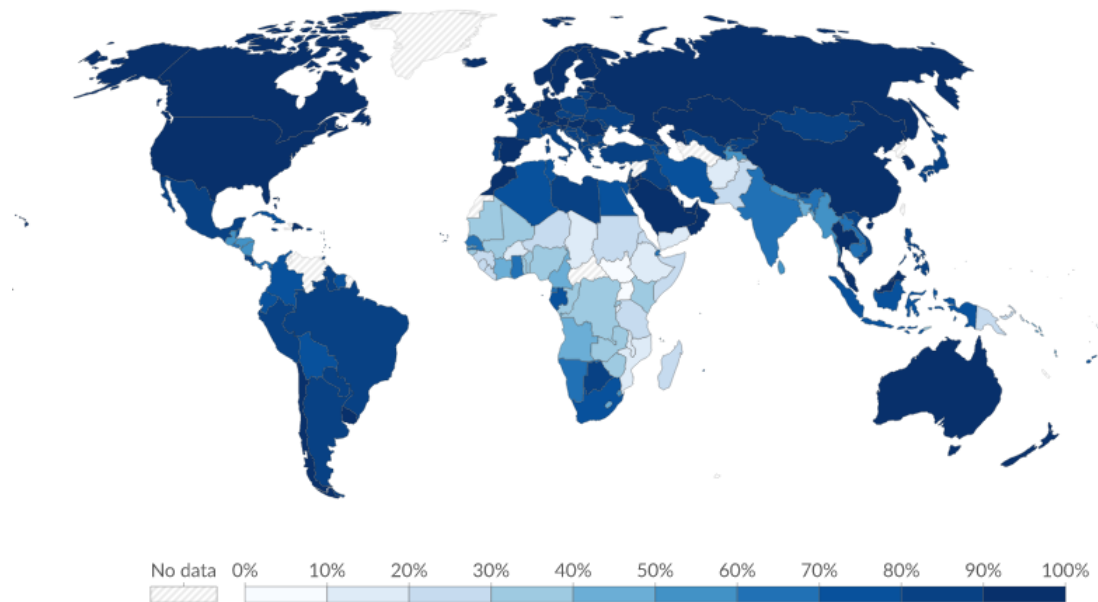
Figure 1

Share of the population using the Internet, 2025.

¹ (International Telecommunication Union [ITU], 2024)

Share of the population using the Internet, 2025

Share of the population who used the Internet in the last three months.



Data source: World Telecommunication/ICT Indicators Database - International Telecommunication Union (ITU), via World Bank (2026)
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Note. From Our World in Data, based on the ITU World Telecommunication/ICT Indicators Database via the World Bank (2026).

Past the connectivity floor, the second layer of exclusion is geography. Stanford's Institute for Human-Centered AI reports that the development of frontier AI models is geographically concentrated in a small number of North American, European, and MENA economies, with the US producing the majority of large-scale AI systems each year and the African continent producing effectively none². The cumulative pattern is presented in Figure 2. The implication for Rwanda is that even where connectivity exists, AI use depends on remote application programming interfaces operated by firms in the Global-North — which introduces cost, latency, and dependency that constrain what schools and ministries can realistically deploy and at what scale.

² (Maslej et al., 2024)

On a third level of analysis, another layer of exclusion is language. Most state-of-the-art models perform best in English and a handful of high-resource European and Asian languages, while Kinyarwanda is what AI linguists call a low-resource language. This intersects with Rwanda's own educational policy in a specific way. Since 2008, the country's official language of instruction has been English, replacing French in most subjects from upper-primary onward, even while Kinyarwanda remains the language most students speak at home. AI tutoring, when it finally arrives, will therefore meet a student from Gacundezi already learning in a second language; its utility for foundational comprehension in lower-primary, where instruction is meant to be in Kinyarwanda, will be sharply limited. Birhane (2021) anchors this point conceptually. She argues for what she calls a "relational ethics" approach in which AI systems are evaluated not as isolated models but as objects entangled with the social and historical structures they operate inside. The absence of Kinyarwanda from training data, in her view, is not a bug to be patched but it is a structural feature of how data sets are built.

Specific Rwandan cases illustrate the layered outcome. The country's One Laptop Per Child rollout in the late 2000s distributed 274,073 XO laptops to 1,624 schools, but evaluation studies in the years that followed found that hardware access alone did not produce learning gains in absence of trained teachers, electricity, and curriculum... The pattern is familiar to anyone following AI in education: a technology described as universal arrives universally, but its benefits rarely do. Higher up the system, institutions like CMU Africa and the AIMS do train Rwandan students in advanced AI methods, but their reach is narrow and their compute infrastructure depends on partnerships with Global-North institutions. The layered access gap therefore reproduces itself at every educational level, from foundational literacy upward.

Policy Instruments of Access

If the divide is layered, one might successfully posit that so are the policies that produce it. Four levels of policy converge on the question of who gets access to frontier AI: international, regional, national, and corporate.

At the international level, the most consequential instrument is the Recommendation on the Ethics of AI, adopted by all 193 UNESCO Member States in November 2021³. It is the first global standard-setting instrument on AI ethics and the first to frame equitable access to AI as a human-rights concern rather than a market-efficiency one. The Recommendation lays out policy action areas in data governance, education, gender, and environment, and calls on Member States to report on implementation. As a Recommendation rather than a treaty, however, *it is soft law*: it sets a normative ceiling and creates reporting obligations, but it cannot compel signatories to take action.

³ (UNESCO, 2021)

At the regional level, the AU⁴ endorsed a Continental AI Strategy in July 2024. The strategy positions African states as co-shapers of AI development rather than passive recipients and identifies coordinated infrastructure investment, talent development, and harmonized regulation as priority areas. While it does not directly deploy computer or train models, it provides political cover, coordination, and bargaining power for individual member states in their respective regional integrations.

Yet, it is at the national level that most action is being taken. Rwanda, for instance, through MINICT released the National AI Policy in 2023 - one of the first such documents on the continent. The policy frames AI as a development lever and commits the state to a national AI skills program, a multi-sectoral data governance task force, and a Rwanda AI Office to drive implementation. The policy's stated mission⁵ is "to leverage AI to power economic growth, improve quality of life and position Rwanda as a global innovator for responsible and inclusive AI". These are concrete institutional instruments, auditable against actual progress as we slowly advance towards AI actualisation and scale.

A fourth, less formal but no less binding level of policy is set in the Global North outside any treaty framework. US export controls on advanced AI chips, since 2022, restricts the flow of high-end compute to a broad set of countries. Closed-weight frontier laboratories control who can run their models and under what terms. LLM application programming interfaces are priced and rate-limited in ways that bind⁶ low-income institutional users. While these don't appear in any UN Recommendation, they still determine more in AI access in practice than any treaty does.

Ultimately, whether - in a market-led view - the private sector should set the terms of AI access or, - in policy - AI should be treated as public infrastructure subject to the human-rights; is still a question up for debate.

Can a Global-South-Led Strategy Close the Gap?

The last question is the one on which the literature is most actively divided. There's two camps in the recent scholarship, and a third, synthesizing the first two, is starting to take shape.

⁴ (AU, 2024)

⁵ (MINICT, 2023)

⁶ (Maslej et al., 2024).

The first camp argues that a Global-South-led AI strategy can, in fact, meaningfully close the gap by reframing AI development around local priorities, data, and talent. Mohamed, Png, and Isaac (2020) import decolonial theory into AI development and propose practical tools, including community participation, reverse tutelage, and the renewal of affective and political communities, for designing AI systems that don't simply replay extractive patterns at machine speed. Their framework gives Global-South actors both a critique and a method, and it furnishes the vocabulary of “algorithmic coloniality⁷” that other scholars including Birhane (2021), Ricaurte (2019), and Oyedemi (2019) have since deployed in African and Latin American contexts.

Applied to Rwanda, this approach suggests that local fine-tuning of open-weight models, community-curated Kinyarwanda data sets, and a national AI Office with implementation authority aren't the old-normal of symbolic political promises but operationally significant.

The second camp argues that meaningful access depends on cooperation with, and concessions from, Global-North actors because the material substrate of AI is concentrated outside the South. Crawford (2021), in the opening of *Atlas of AI*, traces this substrate. She writes that “AI is neither artificial nor intelligent. Rather, AI is both embodied and material, made from natural resources, fuel, human labor, infrastructures, logistics, histories, and classifications” (p. 8). Her catalogue of those material inputs, including the lithium and the cobalt, the fresh water for cooling, the chips, the data centers, and the human labor that labels training data, shows that all of them are physically concentrated in a small set of Global-North geographies. Taken into context, the sovereignty rhetoric moves neither data centers nor the talent pool.

Figure 2

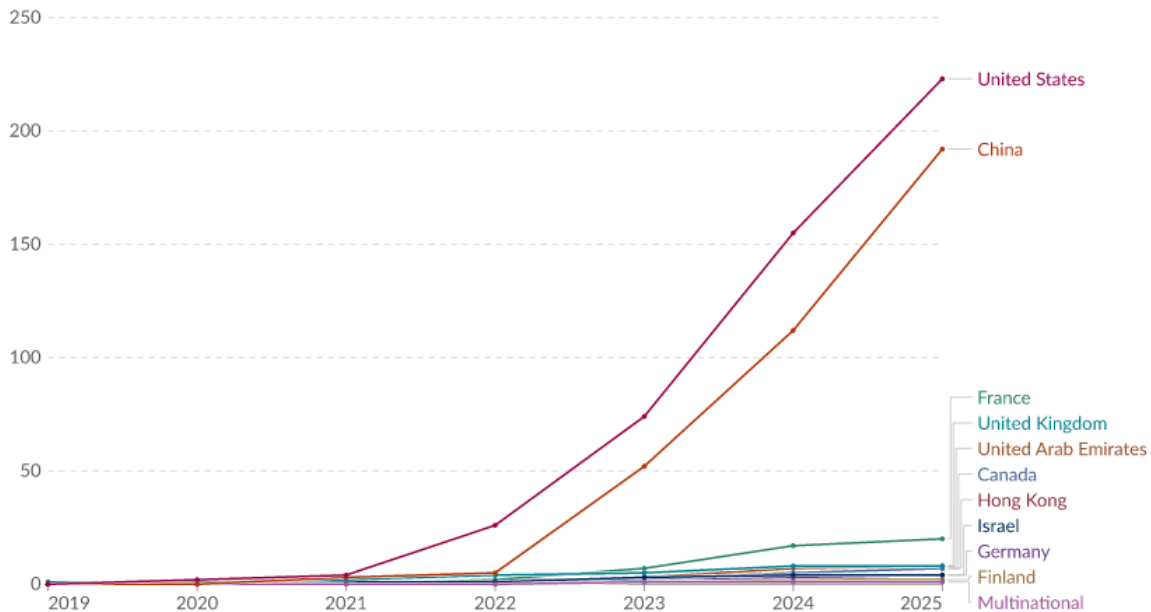
Cumulative number of large-scale AI systems by country since 2019.

⁷ Algorithmic dependence and digital colonialism: a conceptual framework for AI in education and knowledge systems of the Global South.

Cumulative number of large-scale AI systems by country since 2019

Our World
in Data

Refers to the location of the primary organization with which the authors of a large-scale AI systems are affiliated.



Data source: Epoch AI (2026)

OurWorldinData.org/artificial-intelligence | CC BY

Note: The source defines AI models as "large-scale" when their training compute is confirmed to exceed 10^{23} floating-point operations.

Note. From Our World in Data, based on Epoch AI (2026).

Rwanda's own National AI Policy implicitly recognizes this dependency. The AI Policy was developed in partnership with GIZ FAIR Forward, the Centre for the Fourth Industrial Revolution Rwanda, and The Future Society - only positing that policy gets made under conditions of asymmetric capacity.

The third camp, increasingly visible in the recent literature, treats the either-or as a false binary. The hybrid model is a Global-South-led strategy paired with selective, terms-set cooperation rather than passive consumption. The current AI ecosystem includes a growing set of open-weight frontier models, distillation techniques, and local fine-tuning workflows that lower the barrier to building Kinyarwanda-capable systems on top of existing foundations.⁸ Open weights are not a panacea, since they still require compute to fine-tune, but they shift the leverage point from "please give us access" to "we will build on top of what is open." Birhane (2021) cautions, however, that projects framed as AI for Africa can replicate extractive patterns rather than reduce them when the terms of partnership are set by Global-North actors. The hybrid model only works if the Global-South party sets the terms.

Final Words

⁸ (Maslej et al., 2024)

This essay engaged the AI divide in the Global South using Rwanda as its case study and the education system as the point of analysis. On the first research question, the evidence suggests that AI access shapes educational outcomes in Rwanda through a stacked sequence of exclusions: connectivity below the Global-North baseline, near-zero domestic compute, a mismatch between English-language frontier models and a Kinyarwanda-speaking student body, and a hardware-first history of educational technology rollouts whose benefits did not arrive evenly. On the second, the policy levers that determine access operate at four levels, international, regional, national, and corporate, and these levels do not always pull in the same direction. On the third, the evidence supports neither pure autonomy nor passive cooperation. It supports a hybrid model in which a Global-South-led strategy is paired with selective cooperation on terms the Global South sets.

However, the case study has limits worth naming. The evidence is single-country and the policy landscape is moving quickly. School-level outcome data on AI-enhanced learning in Rwanda specifically remains thin, and this essay relies on adjacent indicators (connectivity, model availability, language coverage) as proxies for what cannot yet be measured directly. Future research would benefit from disaggregated school-level data on AI tool usage in Rwandan primary and secondary classrooms, and from comparative work across multiple African states implementing national AI strategies in parallel.

The big picture is that AI access in 2026 is no longer a development question but - increasingly - that of fundamental human-rights. The vocabulary already exists in the UNESCO Recommendation. What is missing is the political will to apply it consistently across the four levels of policy this essay has engaged.

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